Epeirogenic uplift may be itself a contributing cause of widespread glaciation—this would link the original idea of Peterman *et al.* with the hypothesis proposed here. It is not necessary, however, to invoke epeirogenic uplift as a cause of glaciation—it may be almost entirely a consequence of the geographic distribution of the continents (in other words, the concentration of land around the poles results in glaciation)[2].

If the proposed hypothesis is correct, it is possible to predict another maximum in $^{87}Sr/^{87}Sr$ just prior to the beginning of the Cambrian. Volcanism and uplift are possible reasons why the $^{87}Sr/^{86}Sr$ input of the oceans may change, but glacial erosion and sediment transport may be of equal or even greater importance. It might even be possible to argue that glacial activity is the dominant effect and that the isotopic composition curve represents a measure of the intensity of glaciation in past time, superimposed on a general increase of ^{87}Sr in rocks through geologic time (the slope of the long term change due to the increasing age of the Earth is superimposed on Fig. 1). It would be brash, however, to consider the problem solved. Accumulation of high precision Sr isotope data, more exact knowledge of palaeogeography and palaeoclimate, and measures of rates of tectonic activity and rates of generation of volcanic rocks will refine and revise our models for change in sea water Sr isotopic composition. The goal will be a quantitative relationship expressing the significant causes and their effects.

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Early Tertiary Geomagnetic Polarity Reversals in India

THE Deccan Trap flood basalts of peninsular India are believed to be early Tertiary formations, although the age, duration and the mode of eruption of this volcanism are still debated[1–3]. Earlier palaeomagnetic investigations have led to their classification into only two polarity zones[4] (that is, one geomagnetic field reversal) with the elevation of 2,000 ± 200 feet m.s.l. dividing the upper normal (N) and the lower reversed (R) horizons. Inversions of the geomagnetic polarity, however, are known to have been frequent during the late Mesozoic and the Tertiary[5], and this implies that several geomagnetic field inversions occurred during these lava extrusions. Pal[6] has suggested, on the basis of some recent observations, that further intensive palaeomagnetic studies on these lava sequences are necessary to establish evidence of such reversals.

We have made a palaeomagnetic survey of the Deccan Trap flows in Malwa Plateau (22° 30′ N, 75° 47′ E); the fifteen flows were sampled at elevations between 1,060 and 2,800 feet m.s.l. The bottom two flows in these lavas are N, whereas the upper flows can all be grouped together as R (Table 1). (These data were obtained after intensive alternating field demagnetization tests had proved the directional stability of the specimens.)

The lavas in the Malwa Plateau belong to the "simple flows" in the Deccan Traps[7,8] (that is, the individual lavas in these traps had long extrusion times). The presence of as much as 10–20 feet of red bole horizons between these lavas suggests that there were significant time intervals between the extrusions of the successive flows. These lavas are considered geologically to belong to the "Middle Traps" horizon[9] and continue as far north-east as Sagar (23° 56′ N, 78° 38′ E). The age of the base of the Sagar flows has been estimated to be about 60 m.y. by K–Ar methods[10]. The flows of the Malwa Plateau are probably the same age.

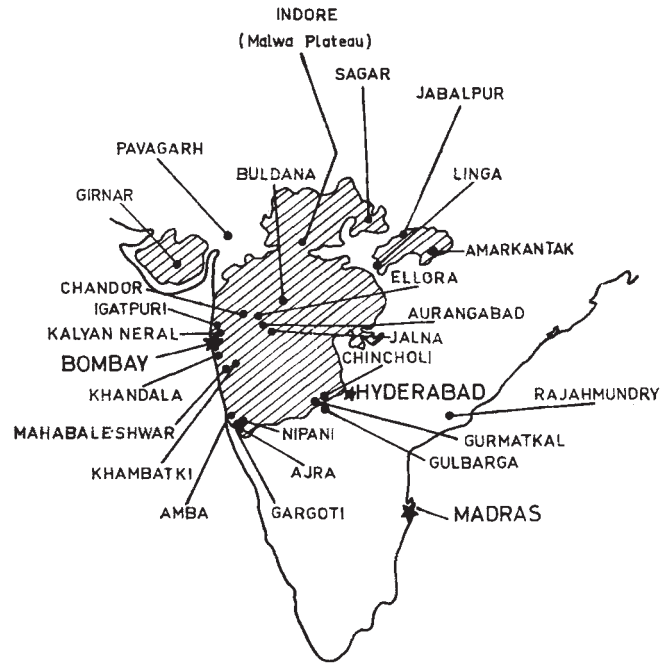


Fig. 1 The sites of palaeomagnetic studies of Deccan Traps.

If the polarity reversal in these lavas is indeed due to reversal of the geomagnetic field polarity, then there is a direct contradiction with the inferences drawn from earlier studies. One possible explanation for the observed polarity sequence in the Traps of Malwa Plateau could be their contemporaneity with the Rajmahal lavas[11] and the Tirupathi sandstones[12]. The Rajmahals are known from palaeontological[13] and K–Ar age data (McElhinny, personal communication) to be N Cretaceous formations and they can therefore be correlated with the Tirupathis (Aptian) where the lower N horizon is overlain by R horizon. Because of the lower Tertiary age of the Deccan Traps of Malwa Plateau the N horizon in these lavas can perhaps be correlated with the N Rajahmundry Traps[14] which are of Lower Eocene age[15]. A correlation is made in Table 2 between the elevations and the remanent magnetic directions of Deccan Trap flows studied in different areas (Fig. 1). Correspondence between elevations and polarity is not observed all over the Trappean area.

The lava sequences studied in the neighbourhood of Aurangabad[16] suggested the presence of several short term

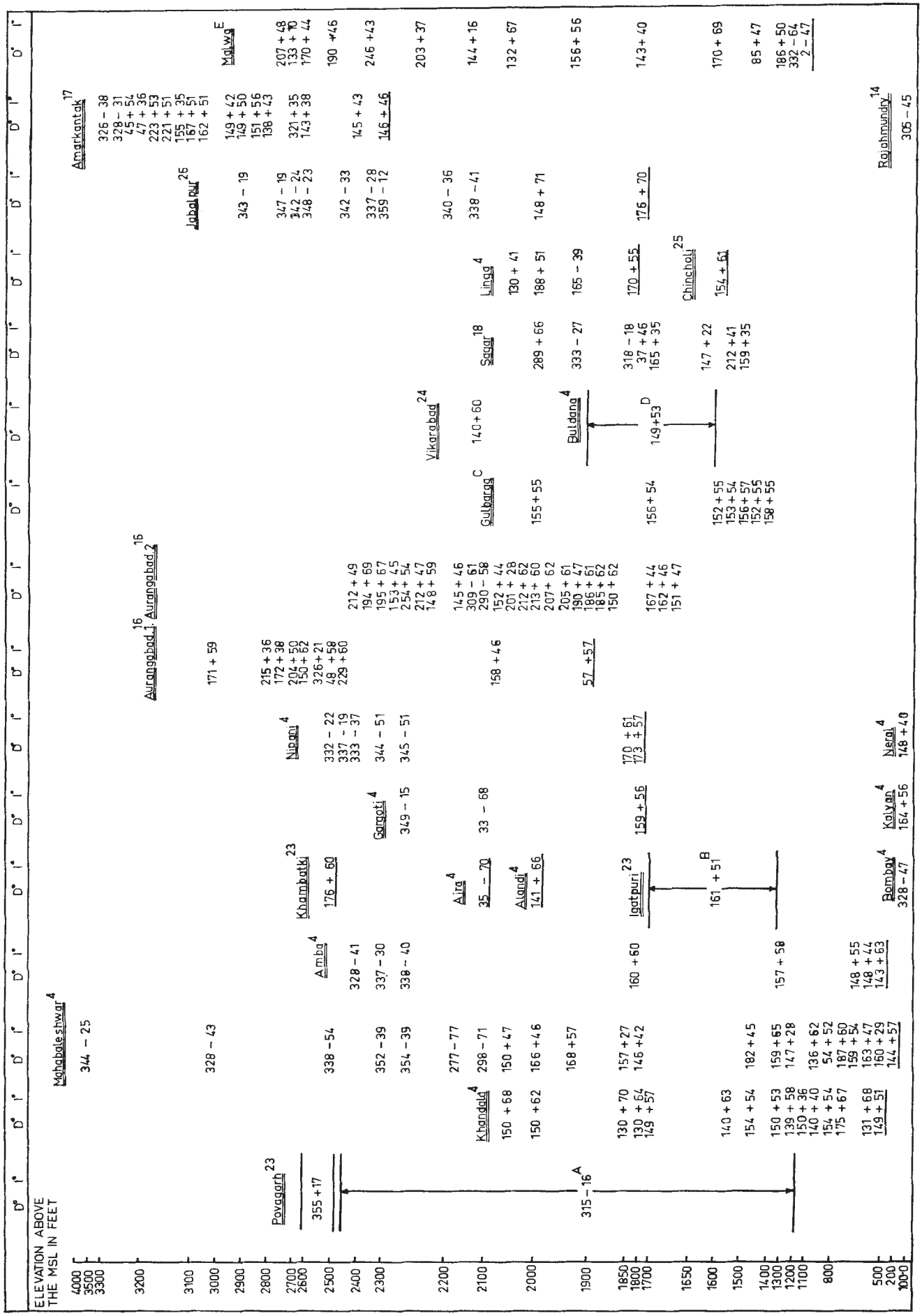
Table 1 Mean Palaeomagnetic Directions of Malwa Plateau Lavas

Elevation above m.s.l. (feet)	No. of flows	No. of samples	Mean declination (east of true north)	Mean inclination (+ve downwards)	Radius of 95% circle of confidence	Polarity group
2,800–1,300	11	59	164°	+49°	8.2°	Reversed *
1,280–1,060	2	17	353°	−54°	7.5°	Normal

* Two flows, at elevations of 2,350 and 1,440 feet, have been omitted because they showed discordant directions, declination 246°, inclination +43°; and declination 85°, inclination +47° respectively.

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Table 2 Flow-Mean Palaeomagnetic Directions of Deccan Trap Lava Sequences



A, MEAN OF 8 FLOWS; B, MEAN OF 3 FLOWS; C, UNPUBLISHED DATA; D, MEAN OF 2 FLOWS; E, PRESENT WORK.

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polarity events and this was confirmed by one of the flows studied in the Amarkantak area[17] (declination 321°, inclination +35°) sandwiched between two R lavas at 2,710 feet. In another area (Sagar) there are indications of one more inversion of the geomagnetic field during the Trappean eruptions[18]. In the Girnar Hills (Saurashtra Peninsula) the boundary between lower R and upper N horizons is at 360 feet m.s.l. and is overlain by two possible R flows at 3,200–3,400 feet (Mittal, G. S., personal communication). In Malwa Plateau the polarity sequence (R horizon being overlain by N horizon) is altogether reversed.

These observations raise the question of short term geomagnetic polarity events as distinct from the geomagnetic polarity epochs. The lavas in the Deccan Traps can be classified into two types[7–8] and although the "compound flows" chiefly occupy the Western Ghats and the Marathawada region of Maharashtra State (the neighbourhood of Aurangabad), the simple flows cover the rest of the Trap country. The former are a result of extrusions in quick succession but the latter are formed over longer periods of time. This makes it difficult to distinguish between polarity events and epochs, because the probability of recording short term polarity events is greater in compound flows. The polarity horizons probably represent epochs due to the time gaps between successive flows in the "simple" case.

It is not clear from geological evidence exactly where the eruptions in the Traps first commenced[1,2]. Pal and Bhimasankaram[19] have, however, shown from palaeomagnetic data on the Deccan lavas that the foci of the volcanism were widely distributed. Athavale and Verma[20] have concluded from the palaeomagnetism of some Gondwana dykes that the Rajmahal and Sylhet lavas and the Deccan lavas were phases of a continuous volcanic activity. The entire volcanism probably commenced during the Cretaceous (100–120 m.y.) and continued into the Tertiary (40–60 m.y.). It is not possible at present therefore to distinguish clearly those lavas in the Trap country belonging to the older phase from those belonging to the younger phases of volcanism; neither is it possible to evolve an elevation criterion to demarcate the polarity inversion horizons in the Deccan lavas.

The results from Malwa Plateau further corroborate this opinion. Pal[6] has suggested that although the two polarity-zone model[4] may be valid for the Traps in the Western Ghats, the question of the number of geomagnetic field reversals during the Deccan Trap volcanism is still open. This also implies that the deduction of the duration[21] of volcanism on the basis of the two-zone elevation–polarity model would not lead to a correct picture of the Trappean volcanic activity. Wellman and McElhinny[22] have recently dated some Deccan lavas by the K–Ar method, but because of the complex geology of these lavas, many more radiometric dating checks have to be made before any conclusions can be drawn about the whole time span of the Trappean volcanicity.

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Interactive Gel Networks for Organic Separations

We wish to report a new and systematic basis for approaching the chemical separations of organic compounds. Interactive gel chromatography with a moving liquid phase is defined by three major factors: solvent absorption (imbibition), steric exclusion, and the combined chemical interactions among gel, solvent and solute. The third involves the reversible process:

$$\text{Acceptor} + \text{Donor} \rightleftharpoons \text{Complex} \qquad (1)$$

This can involve localized electron pair donation as in Lewis basicity, hydrogen bonds, or the more diffuse delocalized nature of pi electrons in unsaturated systems. Our preliminary findings with three different gel network structures are described in this report.

The experimental arrangement, in brief, involves the pumping of organic solvent into a 0.6 by 58 cm column containing the gel network in bead form (bead diameters of 10–30 μm). The effluent stream was monitored using low dead volume spectrophotometric (UV) and refractometric detectors[1] obtained from Laboratory Data Control. Under typical conditions an inlet pressure of 50 p.s.i. and a flow rate of 0.5 ml. minute^{-1} produced column efficiencies in the range of 200–600 theoretical plates for 5 mg, or less, of solute. The solute elution volumes (V_e, expressed in ml.) were measured with a relative standard deviation of approximately 5%.

The first gel was prepared using suspension polymerization[2] to obtain poly(2-methyl-5-vinylpyridine/0.08 divinylbenzene), or PVP. The swollen gel imbibed 65% by weight of chloroform solvent, used as the moving phase. The elution volumes for PVP with chloroform are given in Table 1. The sequence is: polystyrene which is sterically excluded, followed by the basic amines and the cyclohexane; these solutes show no obvious affinity for the PVP gel. A larger retention volume with aniline suggests the effect of the two slightly acid hydrogens. Then the alcohols emerge in order of their increasing acidity or degree of hydrogen bonding toward the pyridine nitrogen.

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